# Local OpenClaw x GitHub Copilot Tutorial

Penn State University — Cyber Security Laboratory

# What are OpenClaw and GitHub Copilot?

- **OpenClaw** is an open-source and autonomous AI ecosystem, designed as a personal assistant
  - Instant messaging with popular chat platforms like Discord, Telegram, and Signal!
  - Extend your OpenClaw with dedicated **Skills**, teaching it fine-tuned control of apps like Gmail and Obsidian!
- **GitHub Copilot** is Microsoft's AI-powered coding utility, sharing support for dozens of unique models
  - Integrates with both IDEs like VS Code and standalone applications like OpenCode and OpenClaw!
  - Students have free access to a **GitHub Copilot Pro** subscription with the **GitHub Student Developer Pack**!

OpenClaw

GitHub Copilot

# Why a Virtual Machine and not Docker?

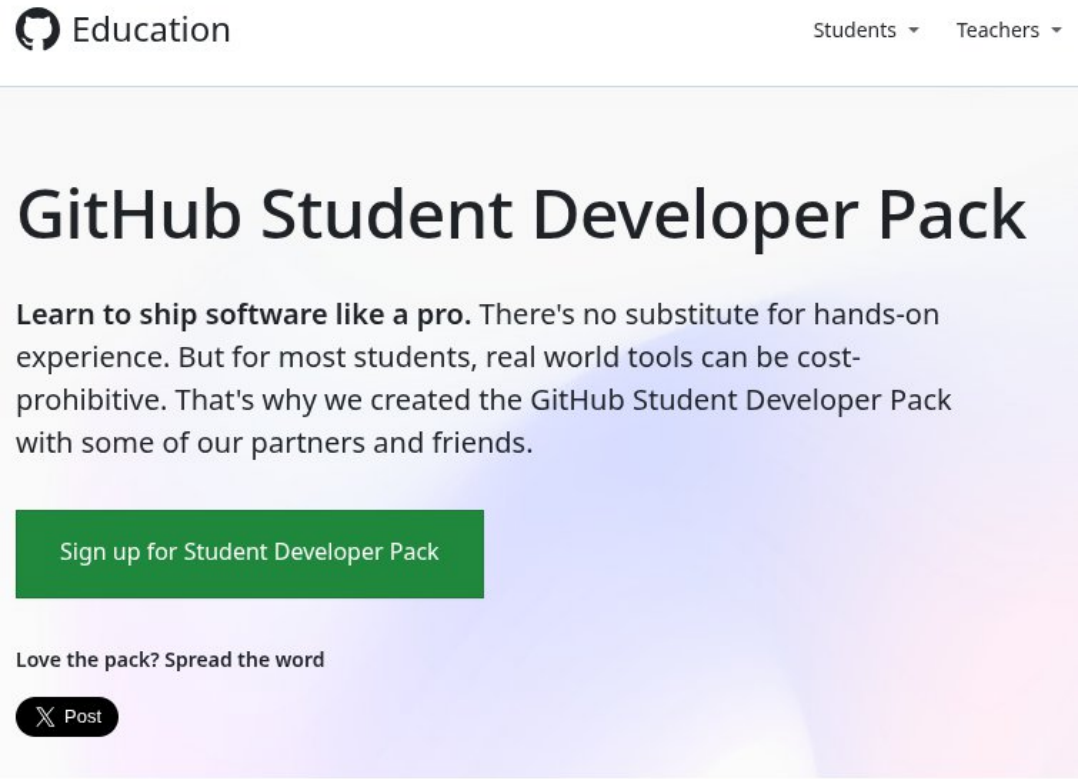
- Many **OpenClaw** features depend on OS components only found in a full Linux installation (i.e., systemd)
  - A Docker container often results in more bugs and unexpected failures during setup.
- Stronger isolation from the host system and its kernel
  - Malicious processes can escape from the Docker socket.
  - A virtual machine only needs an SSH tunnel to the management port 18789.
- **OpenClaw** can only access the documents and files that you give it explicit access to

# Software Requirements

- Git
- Vagrant
- vagrant-sshfs
  - **Only after installing Vagrant**
  - In your terminal, run the command ***vagrant plugin install vagrant-sshfs***:
  - 
```
vagrant plugin install vagrant-sshfs
```
- VirtualBox (Windows, macOS, Linux) or libvirt (Linux)

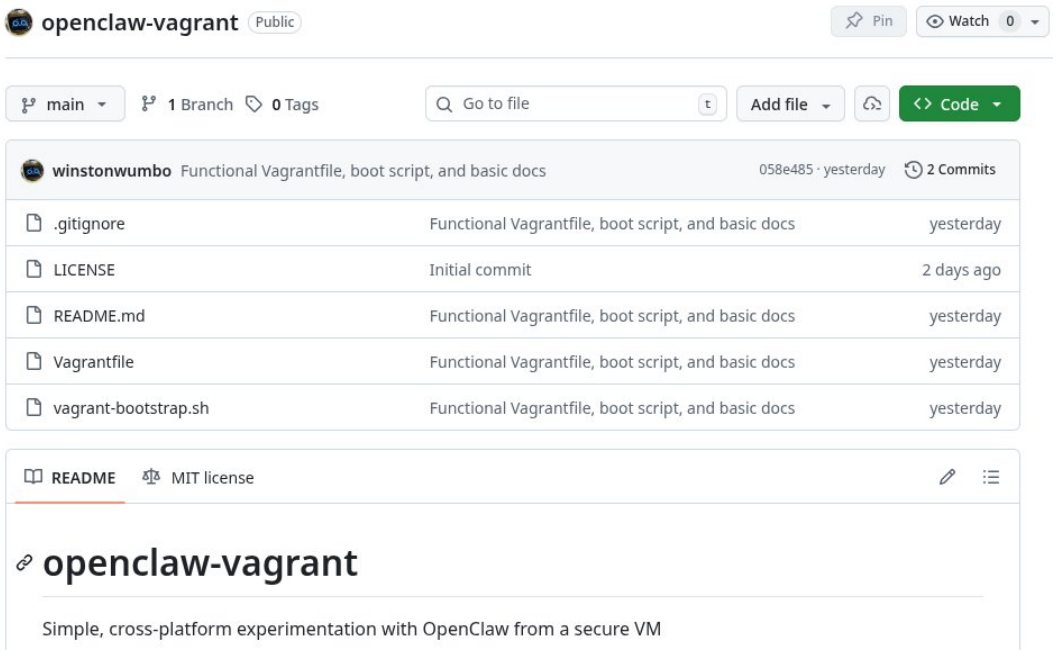
# Requesting Access to GitHub Copilot

- Visit **GitHub Education**, then submit proof of enrollment:
  - https://education.github.com/pack
- Once you receive an approval email, your account is ready!



# Downloading Project Files

- Everything that we need for this **OpenClaw Tutorial** is stored at:
  - [https://github.com/winstonwumbo/openclaw-vagrant](https://github.com/winstonwumbo/openclaw-vagrant)
- To use the files on your local device:
  - Run the command ***git clone https://github.com/winstonwumbo/openclaw-vagrant.git***
  - 
```
git clone https://github.com/winstonwumbo/openclaw-vagrant.git
```
  - Then run the command ***cd openclaw-vagrant*** to enter the folder
  - 
```
cd openclaw-vagrant/
```



# Setting Up with Vagrant

- The setup process for **OpenClaw** and its dependencies is automated through **Vagrant**
- To start building your virtual machine:
  - Run the command ***vagrant up***:
  - 
```
vagrant up
```

- Once **Vagrant** is finished building the virtual machine, it will return control of the terminal to you.

```
ruyu  …/openclaw-vagrant  main  vagrant up
Bringing machine 'default' up with 'libvirt' provider...
==> default: Checking if box 'generic/ubuntu2204' version '4.3.12'
...
[fog][WARNING] Unrecognized arguments: libvirt_ip_command
==> default: Creating image (snapshot of base box volume).
==> default: Creating domain with the following settings...
==> default:  -- Name:              openclaw-vagrant_default
==> default:  -- Description:       Source: /var/home/ruyu/Workspa
enclaw-vagrant/Vagrantfile
==> default:  -- Domain type:       kvm
==> default:  -- Cpus:              2
==> default:  -- Feature:           acpi
==> default:  -- Feature:           apic
==> default:  -- Feature:           pae
==> default:  -- Clock offset:      utc
==> default:  -- Memory:            4096M
```

# Initializing OpenClaw

- Enter your virtual machine with the command ***vagrant ssh***:
  - 
```
vagrant ssh
```

- Initialize **OpenClaw** with the command ***openclaw onboard --install-daemon***:
  - 
```
openclaw onboard --install-daemon
```

# OpenClaw Onboarding: Step-by-Step

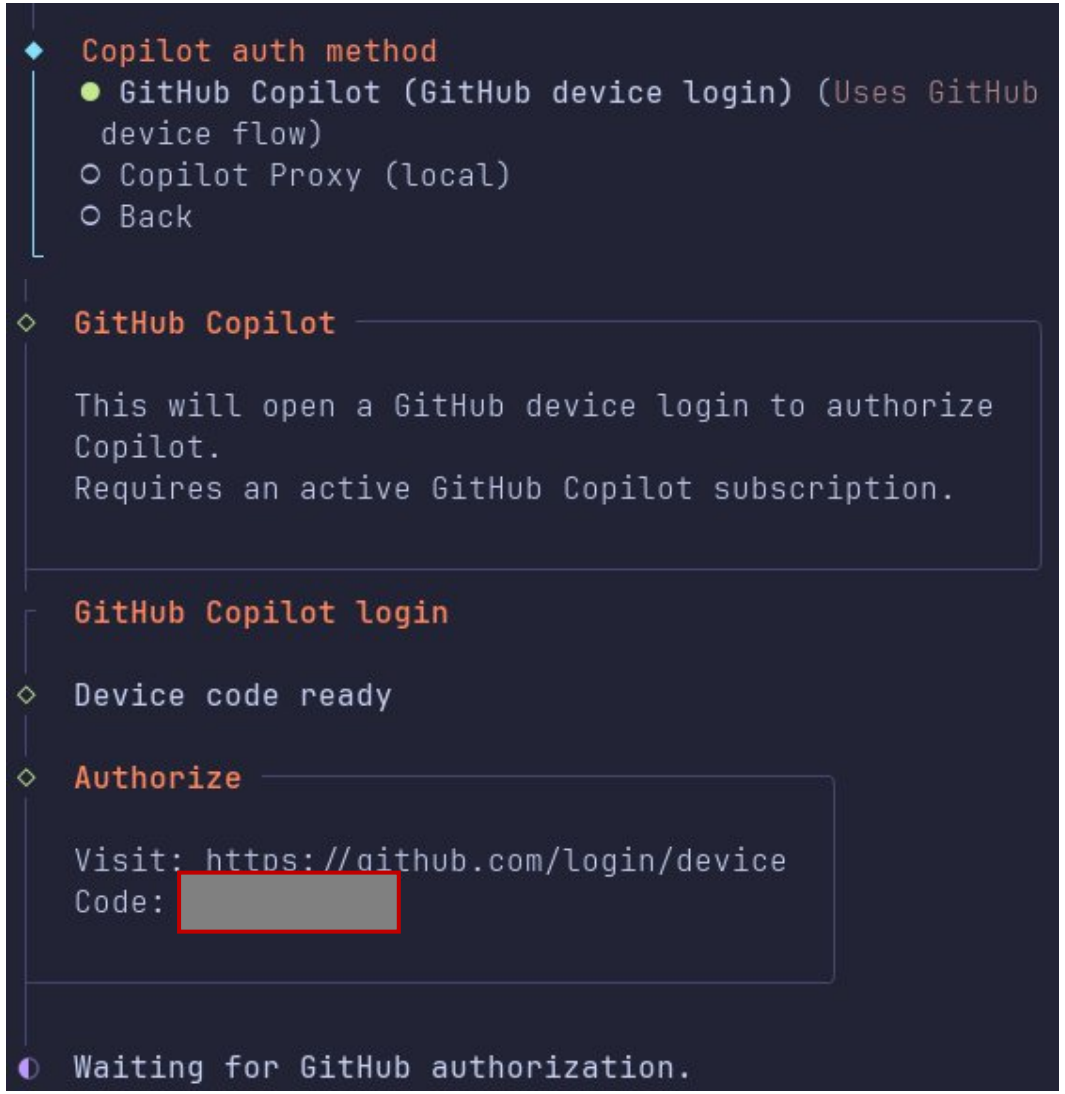
- Select **Yes**
- Select **QuickStart**
- Select **Copilot (GitHub + local proxy)**

```
◇  I understand this is personal-by-default and
shared/multi-user use requires lock-down. Continue?
   Yes

◆  Onboarding mode
   ● QuickStart (Configure details later via openclaw
   configure.)
   ○ Manual

◆  Model/auth provider
   ○ OpenAI
   ○ Anthropic
   ○ Chutes
   ○ vLLM
   ○ MiniMax
   ○ Moonshot AI (Kimi K2.5)
   ○ Google
   ○ xAI (Grok)
   ○ Mistral AI
   ○ Volcano Engine
   ○ BytePlus
   ○ OpenRouter
   ○ Kilo Gateway
   ○ Qwen
   ○ Z.AI
   ○ Qianfan
   ● Copilot (GitHub + local proxy)
   ○ Vercel AI Gateway
```

# OpenClaw Onboarding: Step-by-Step

- Select **GitHub Copilot (GitHub device login) (Uses GitHub device flow)**

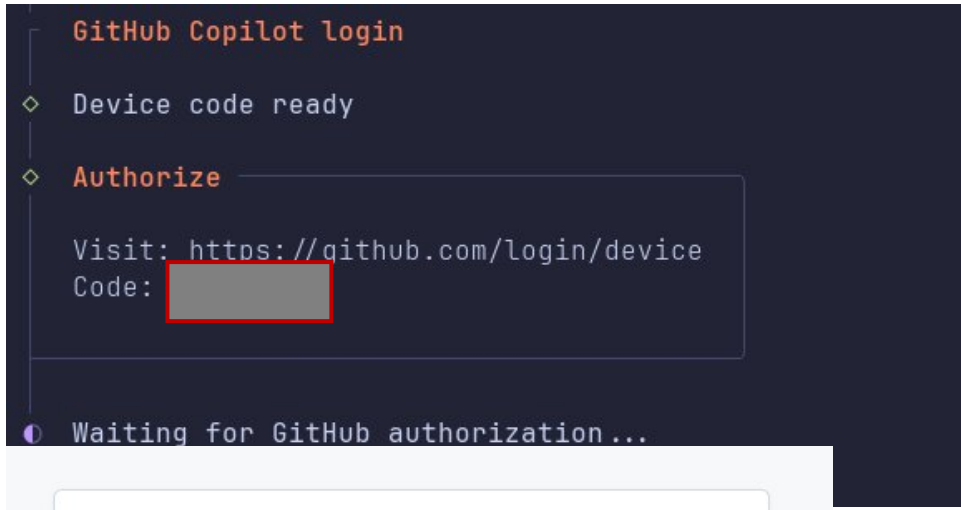
- Open the following **GitHub** link in your browser:
  - https://github.com/login/device

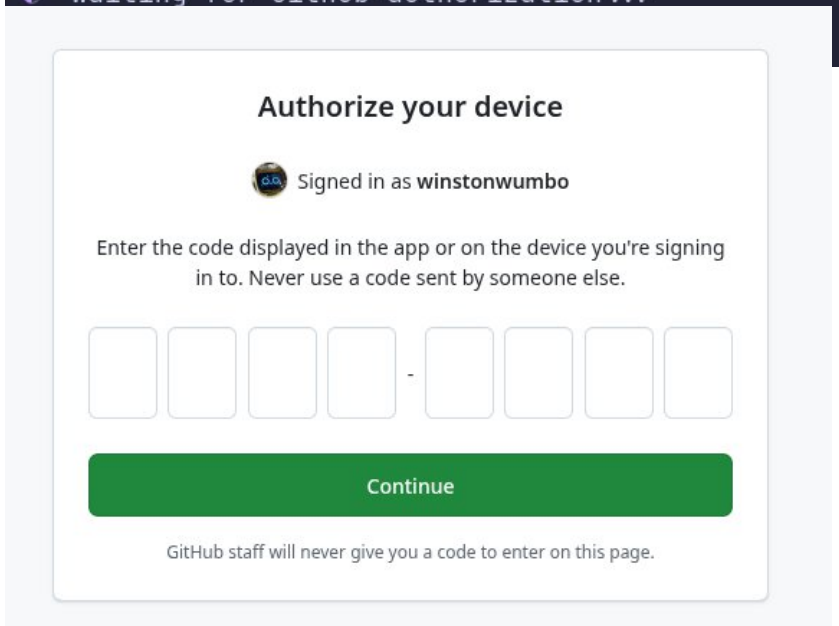


# OpenClaw Onboarding: Step-by-Step

- Copy the authentication code that appears under:
  - **Authorize -> Code**

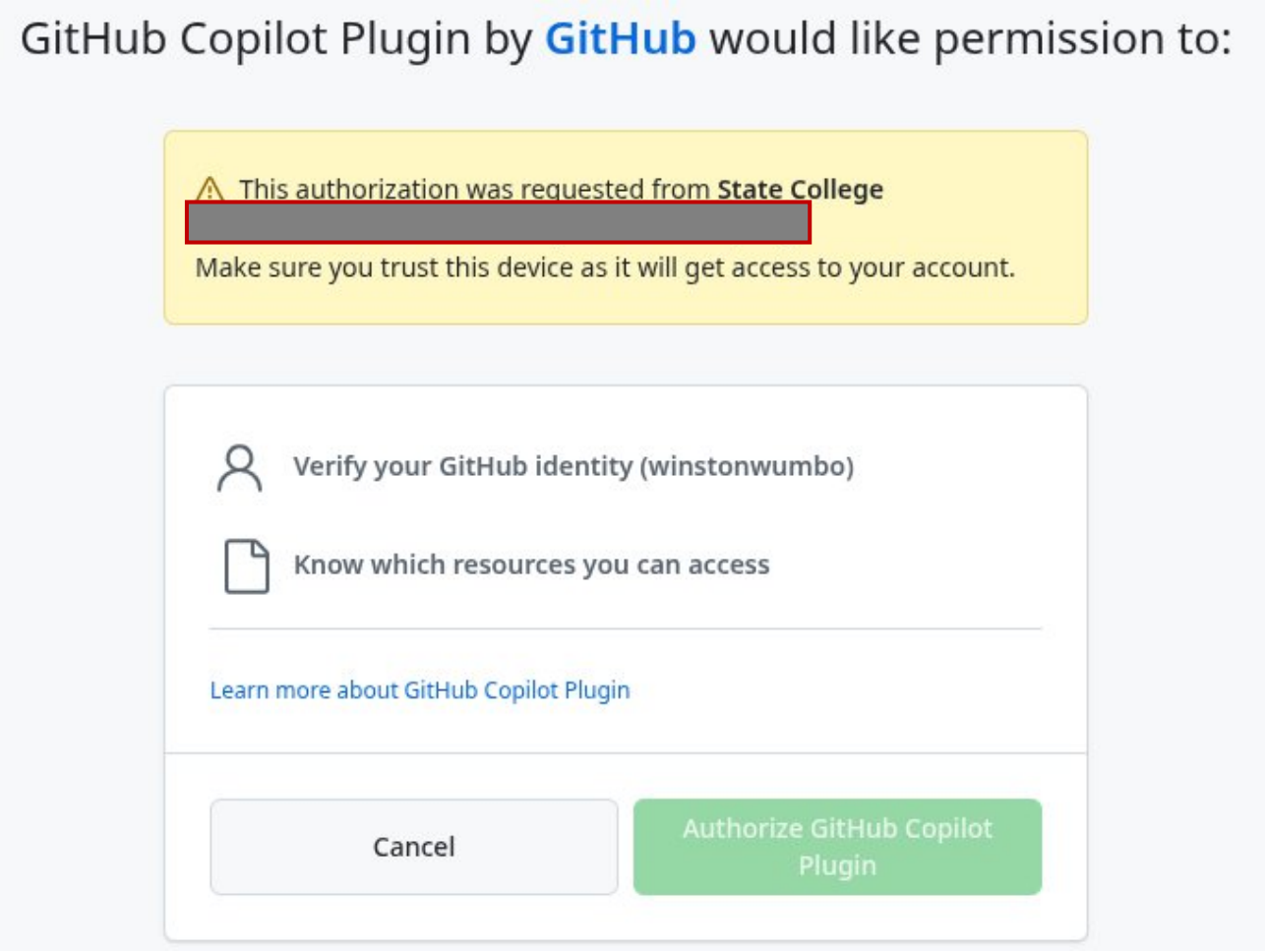
- Enter the authentication code using the same **GitHub** link in your browser:
  - https://github.com/login/device



# OpenClaw Onboarding: Step-by-Step

- Select **Authorize GitHub Copilot Plugin**

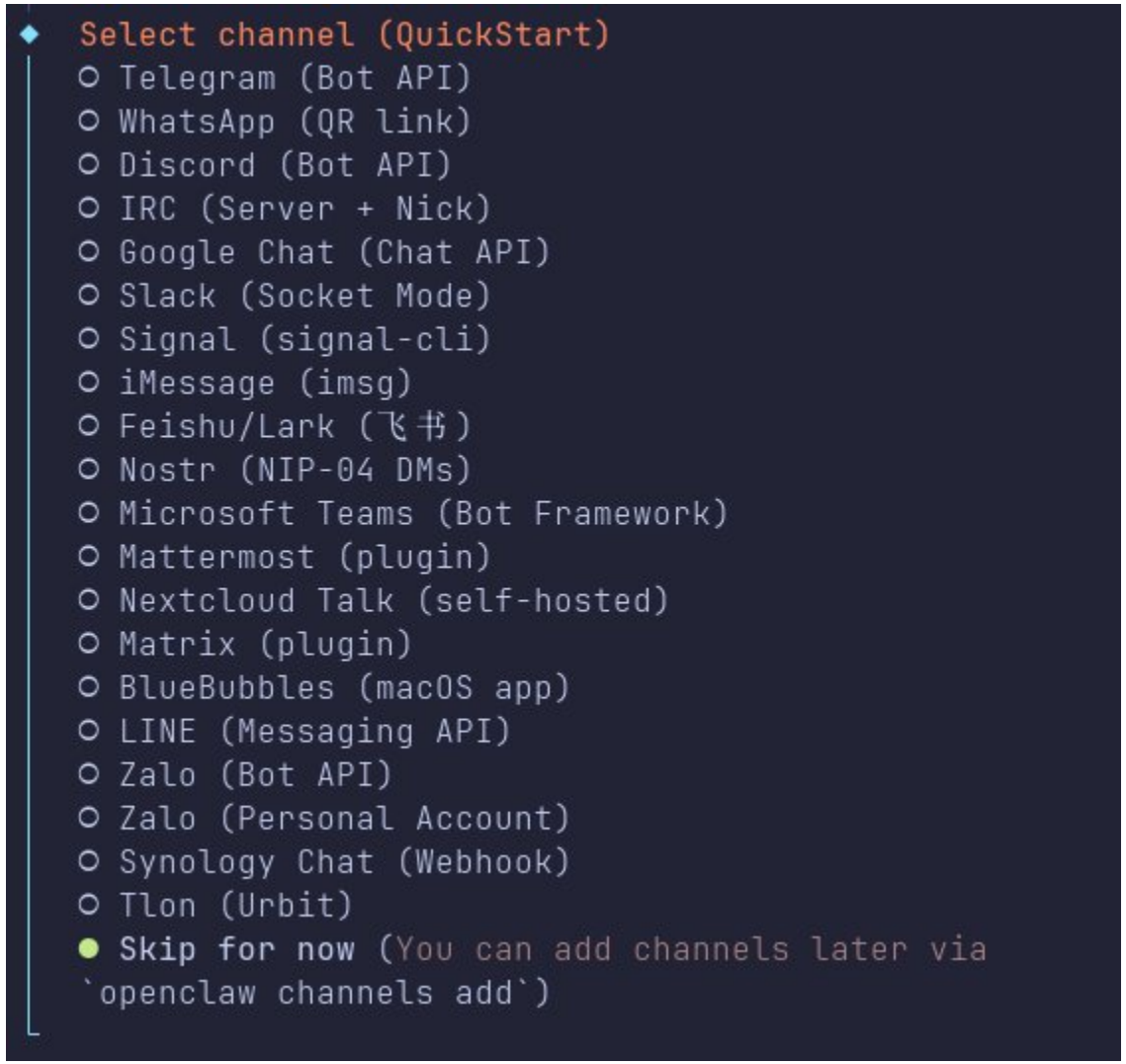


# OpenClaw Onboarding: Step-by-Step

- Select **github-copilot/gpt-5-mini (GPT-5-mini * ctx 125k * reasoning)**
  - GPT-5-mini is a reasonable model with unlimited free requests under GitHub Copilot
- You may select other models based on personal preference after **Onboarding**

# OpenClaw Onboarding: Step-by-Step

- Select **skip for now (You can add channels later via `openclaw channels add`)**
  - We simply interact with OpenClaw through the local dashboard (Web UI)
  - This lets us efficiently validate GitHub Copilot integration is successful
- If adding a channel later, each requires unique configuration for the specific chat platform.
  - This is beyond the scope of the tutorial, but guides are available online.



# OpenClaw Onboarding: Step-by-Step

- Select **No**
  - You can add skills later through the local dashboard (Web UI)
  - Each requires specific credentials (i.e., API key for Brave Search, App Password for Google Workspace, etc.)

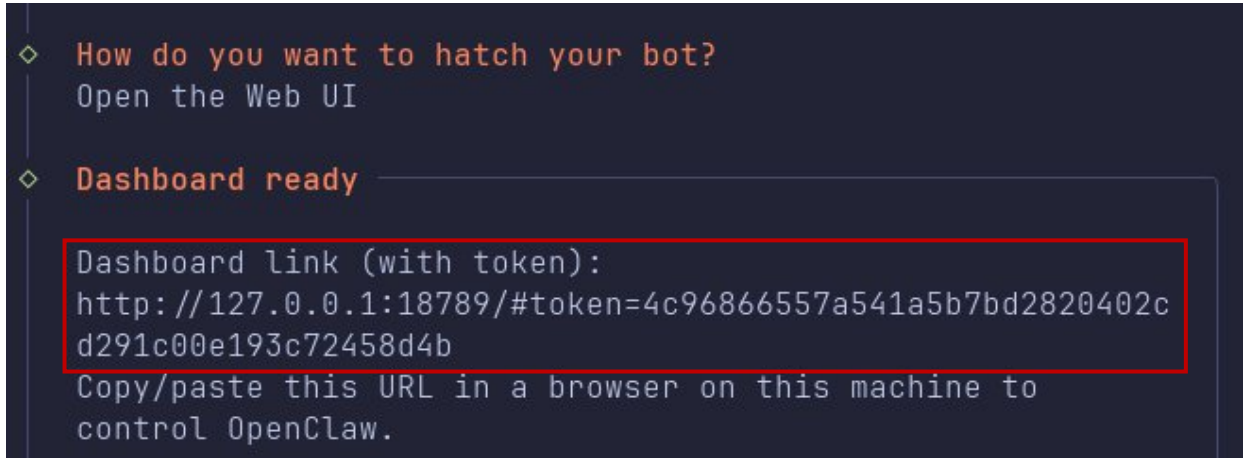
# OpenClaw Onboarding: Step-by-Step

- Select all 4 hook options:
  - **boot-md (Run BOOT.md on gateway startup)**
  - **bootstrap-extra-files (Inject additional workspace bootstrap files via glob/path patterns)**
  - **command-logger (Log all command events to a centralized audit file)**
  - **session-memory (Save session context to memory when /new or /reset command is issued)**
- Each hook provides helpful defaults for **OpenClaw** memory and security

```
◇ Hooks

  Hooks let you automate actions when agent commands are
  issued.
  Example: Save session context to memory when you issue
  /new or /reset.

  Learn more: https://docs.openclaw.ai/automation/hooks


◆ Enable hooks?
  ◻ Skip for now
  ■ 🚀 boot-md (Run BOOT.md on gateway startup)
  ■ 📎 bootstrap-extra-files (Inject additional
  workspace bootstrap files via glob/path patterns)
  ■ 📝 command-logger (Log all command events to a
  centralized audit file)
  ■ 💾 session-memory (Save session context to memory
   when /new or /reset command is issued)
```

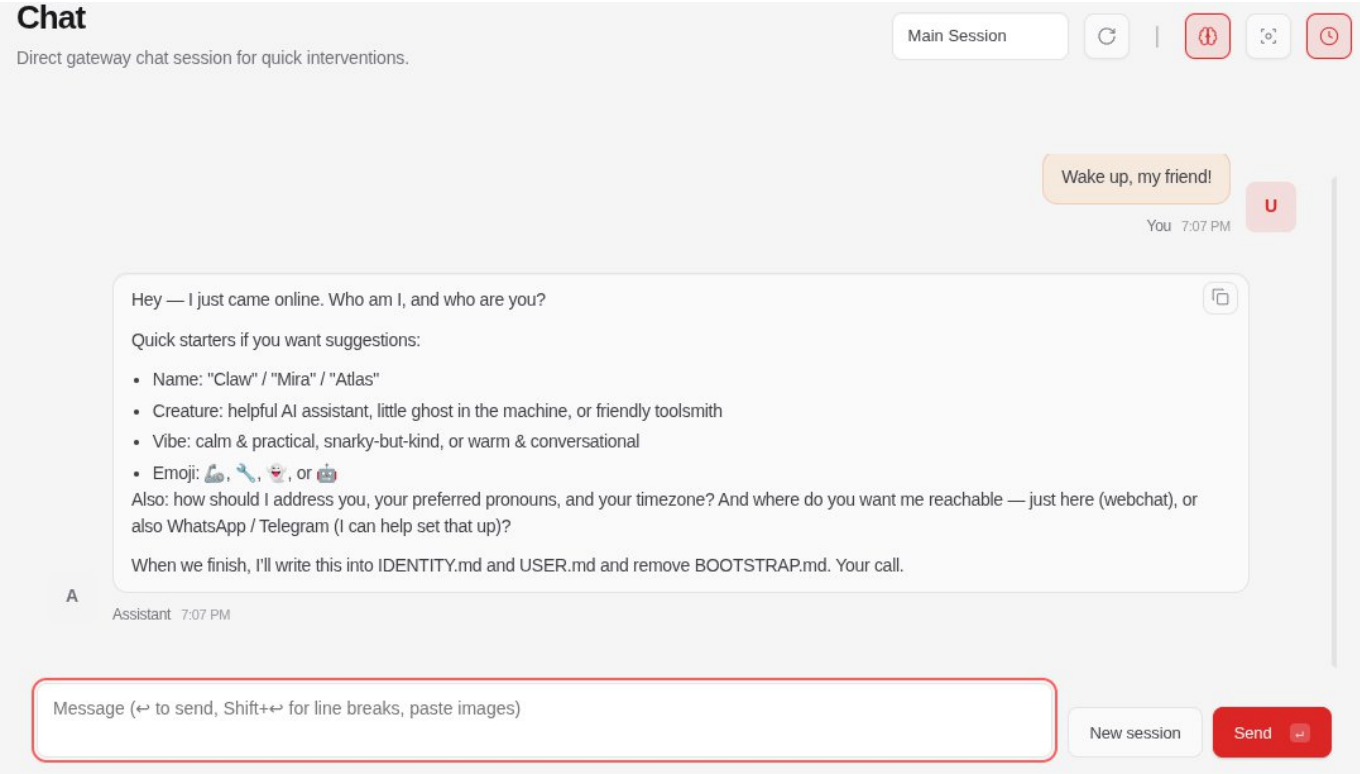
# OpenClaw Onboarding: Step-by-Step

- Select **Open the Web UI**
- You will be presented with a dashboard link with a token:
  - It will look similar to: ***http://127.0.0.1:18789/#token=<token>***
- Open this dashboard link in your browser
- You're now ready to chat with **OpenClaw**!



# OpenClaw Onboarding: Completed!

- Send the message **”Wake up, my friend!”**
- This message puts your **OpenClaw agent** in activation mode
- Introduce yourself!



# Managing the OpenClaw Virtual Machine

- To shutdown the virtual machine:
  - Run ***vagrant halt***
- To suspend the virtual machine (while saving state):
  - Run ***vagrant suspend***
- To turn on the virtual machine:
  - Run ***vagrant up***
- To connect to the virtual machine:
  - Run ***vagrant ssh***
  - This is necessary to activate the network bridge between your device and the virtual machine

```
vagrant halt
```

```
vagrant suspend
```

```
vagrant up
```

```
vagrant ssh
```

# OpenClaw: Troubleshooting

- If you receive an **unauthorized: device token mismatch** error:
  - Enter the command ***openclaw dashboard*** to generate a new token link
  - 
```
vagrant@openclaw-vm:~$ openclaw dashboard

🦞 OpenClaw 2026.3.2 (85377a2) — We ship features faster than Apple ships calculator updates.

Dashboard URL: http://127.0.0.1:18789/#token=4c96866557a541a5b7bd2820402cd291c00e193c72458d4b
```
- If you don’t want to enter an IP address to access your local **OpenClaw Dashboard**:
  - Enter http://localhost:18789/ instead, it resolves to the same address.